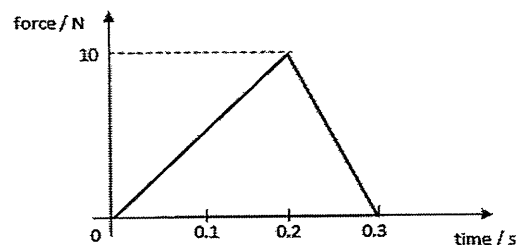


Important note: *denotes questions that may require knowledge from later chapters in order to solve them.

I. MCQs

1. [ACJC/Prelims 2014/P1/6]

A varying force acts upon an object. The graph below shows how the force acting on the object varies with time.



The magnitude of the maximum change in momentum of the object is

- A 3.0 N s B 2.0 N s C 1.5 N s D 1.0 N s

2. [ACJC/Prelims 2014/P1/7]*

A stationary Thoron nucleus of mass $220u$ emits an alpha particle of mass $4u$ with kinetic energy E_α .

Which of the following gives the correct value of the ratio

$$\frac{\text{speed of alpha particle}}{\text{speed of the recoiling daughter nucleus}}$$

and kinetic energy of the daughter nucleus immediately after the emission?

	$\frac{\text{speed of alpha particle}}{\text{speed of the recoiling daughter nucleus}}$	kinetic energy of the recoiling daughter nucleus
A	55	$\frac{1}{55} E_\alpha$
B	54	$(\frac{1}{54})^2 E_\alpha$
C	54	$\frac{1}{54} E_\alpha$
D	$\frac{1}{54}$	$54 E_\alpha$

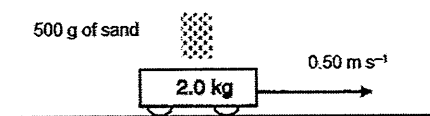
Raffles Institution

Revision Package (H2 Physics)

6 – Collisions

3. [AJC/Prelims 2014/P1/6]

The diagram shows a trolley moving on a frictionless horizontal table at a speed of 0.50 m s^{-1} . 500 g of sand is then released onto the trolley.



What is the change in the momentum of the trolley?

- A zero B 0.15 N s C 0.20 N s D 1.8 N s

4. [CJC/Prelims 2014/P1/7]*

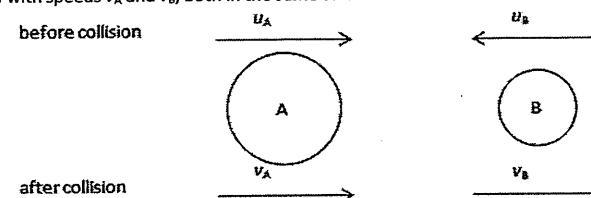
A stationary polonium nucleus ($A = 210$, $Z = 84$) emits an α particle with kinetic energy E_α .

Which of the following is the kinetic energy of the recoiling nucleus?

- A $\frac{2}{105} E_\alpha$ B $\frac{2}{103} E_\alpha$ C $\frac{2}{5} E_\alpha$ D $\frac{42}{103} E_\alpha$

5. [DHS/Prelims 2014/P1/6]

Two spheres A and B approach each other along the same straight line with speed u_A and speed u_B . The spheres collide and move off with speeds v_A and v_B , both in the same direction as the initial direction of sphere A, as shown.



Which equation applies to an elastic collision?

- A $u_A + u_B = v_B - v_A$
 B $u_A - u_B = v_B - v_A$
 C $u_A - u_B = v_B + v_A$
 D $u_A + u_B = v_B + v_A$

6. [HCI/Prelims 2014/P1/7]*

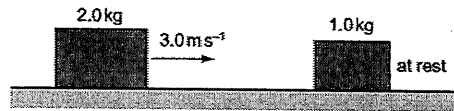
A stationary radioactive nucleus decays into two nuclei, one of which is an alpha particle which has a mass of 4 units and the other, the daughter nucleus of mass 40 units.

If the daughter nucleus has an initial kinetic energy of 200 eV, what is the kinetic energy of the alpha particle?

- A 20 eV B 500 eV C 2000 eV D 20 000 eV

7. [JJC/Prelims 2014/P1/7]

A 2.0 kg mass travelling at 3.0 m s^{-1} on a frictionless surface collides head-on with a stationary 1.0 kg mass. The masses stick together on impact.

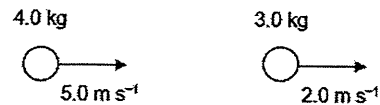


How much energy is lost on impact?

- A 0 J B 2.0 J C 2.4 J D 3.0 J

8. [JJC/Prelims 2014/P1/8]

Two masses moving to the right as shown below make a head-on elastic collision.

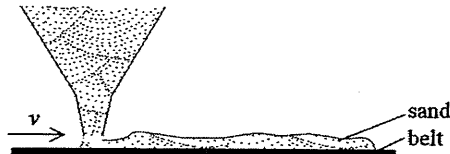


What are the velocities of the two masses after the collision?

	4.0 kg mass	3.0 kg mass
A	1.6 m s^{-1} to the left	4.6 m s^{-1} to the right
B	1.6 m s^{-1} to the right	4.6 m s^{-1} to the left
C	2.4 m s^{-1} to the left	5.4 m s^{-1} to the right
D	2.4 m s^{-1} to the right	5.4 m s^{-1} to the right

9. [NJC/Prelims 2014/P1/5]

Sand falls vertically on a conveyor belt at a rate of $m \text{ kg s}^{-1}$.



In order to keep the belt moving at constant speed v the horizontal force that must be exerted on the belt is

- A mv B $\frac{1}{2}mv$ C mv^2 D $\frac{1}{2}mv^2$

10. [PJC/Prelims 2014/P1/5]

A bomb was projected with a velocity of 100 m s^{-1} at an angle of 40° above the horizontal. At the highest point, it explodes into three equal fragments. One goes vertically upwards with velocity of 50 m s^{-1} , the second moves vertically downwards with velocity of 50 m s^{-1} .

The third fragment moves with a velocity of

- A 77 m s^{-1} horizontally.
 B 230 m s^{-1} horizontally.
 C 77 m s^{-1} at an angle of 40° above the horizontal.
 D 100 m s^{-1} at an angle of 40° above the horizontal.

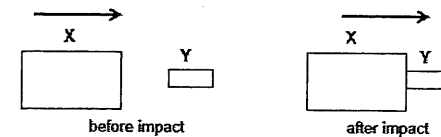
11. [SAJC/Prelims 2014/P1/6]

A body initially at rest explodes into two fragments of mass M and $3M$, having a total kinetic energy of E . The kinetic energy of the fragment of mass M after the explosion is

- A $\frac{E}{4}$ B $\frac{E}{3}$ C $\frac{2E}{3}$ D $\frac{3E}{4}$

12. [SAJC/Prelims 2014/P1/8]

A body X collides with a body Y. The two bodies stick together on impact and then move together.



Which statement is correct?

- A Some of X's momentum becomes kinetic energy of Y
 B Some of X's momentum is lost as heat
 C The momentum of the system is reduced by the forces during impact
 D X transfers some of its momentum to Y

13. [SRJC/Prelims 2014/P1/7]

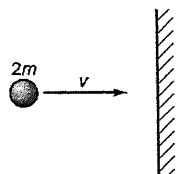
A ball of mass 300 g is being kicked and it leaves a table horizontally with a speed of 10 m s^{-1} . The falling ball then enters a stationary cart of mass 1.2 kg at an angle of 30° below the horizontal. After the ball lands into the cart, the cart immediately moves off with the ball horizontally at a particular speed. Air resistance is considered to be negligible.

What is the final speed and momentum of the ball and cart?

	final speed/ m s^{-1}	final momentum / kg m s^{-1}
A	1.7	2.6
B	2.0	3.0
C	5.8	7.0
D	5.8	8.7

14. [TJC/Prelims 2014/P1/9]

A particle of mass $2m$ and velocity v strikes a wall.



The particle rebounds along the same path after colliding with the wall. The collision is inelastic. What is a possible change in the momentum of the ball during the collision?

- A mv B $2mv$ C $3mv$ D $4mv$

15. [TPJC/Prelims 2014/P1/7]

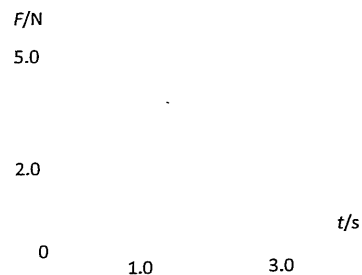
Object X, with kinetic energy E collides with a stationary object Y. Both objects have the same mass. After colliding, X and Y travel onwards together as a single body.

How much kinetic energy is lost in the collision?

- A 0 B $\frac{E}{4}$ C $\frac{E}{2}$ D $\frac{3E}{4}$

16. [VJC/Prelims 2014/P1/6]

The graph below shows how the force acting on a 2.0 kg body varies with time.

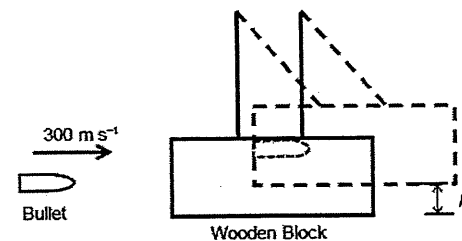


Assuming that the body is initially moving in a straight line at 2.0 m s^{-1} , what is its final velocity?

- A 4.0 m s^{-1} B 6.5 m s^{-1} C 9.0 m s^{-1} D 11 m s^{-1}

17. [YJC/Prelims 2014/P1/10]

A bullet of mass 40.0 g is fired horizontally into a suspended wooden block of mass 2.00 kg with a velocity of 300 m s^{-1} . This is shown in the diagram below. The bullet gets embedded in the block and the bullet and block then move off as one body.



What is the maximum height h attained by the bullet and the block as it swings upwards?

- A 1.76 m B 23.3 m C 40.1 m D 89.9 m

II. Structured Questions

1. [JJC/Prelims 2014/P2/2]

- a. A simple two-stage rocket system consists of two components A and B as shown in Fig. 1.1. They are stacked one on top of another and fired in stages.

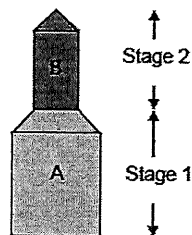


Fig. 1.1

When the two components separate in space, the variation of the force that component A exerts on component B is shown in Fig. 1.2 below.

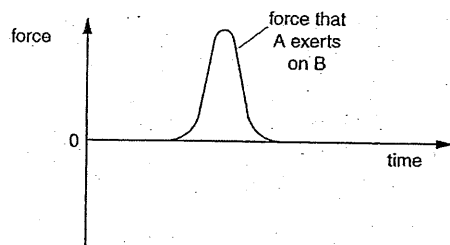


Fig. 1.2

- On Fig. 1.2, sketch the graph of the force that component B exerts on component A. [1]
- Suggest how the answer to i. is consistent with the Principle of Conservation of Linear Momentum. [2]
- The two-stage rocket consists of component A of mass $3m$ and component B of mass $2m$. The rocket is travelling in space at a constant speed V_0 when an internal explosion causes component A to move backwards with a speed of $\frac{1}{3}V_0$. Determine, in terms of V_0 the speed of component B after the internal explosion. [2]
- Explain briefly why this two-stage rocket launching is advantageous compared to a single-stage launching from Earth. [2]

- b. Fig. 1.3 shows speed-time graphs for three spherical objects made of the same material but different radii, a_1 , a_2 , a_3 , falling through a column of liquid after they are released from rest.

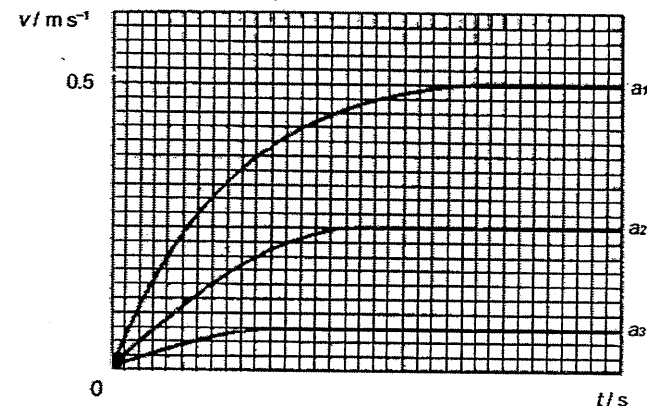


Fig. 1.3

- Explain qualitatively why the speeds tend to a constant value. [3]
- If the constant speed of a falling sphere is proportional to the square of its radius, calculate a_2 given that $a_1 = 2.0$ mm. [2]

[Total: 12]

2. [MJC/Prelims 2014/P2/1]

A large metal ball is hung from a static crane by a cable of length 5.8 m as shown in Fig. 2.1. In order to knock down a wall, a metal ball of mass 380 kg is pulled away from the wall and then released. The ball moves through an arc and makes contact with the wall when the cable is vertical.

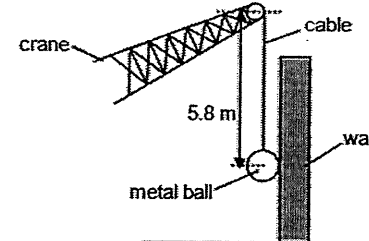


Fig. 2.1

The graph in Fig. 2.2 shows the variation of the speed of the ball v with time t .

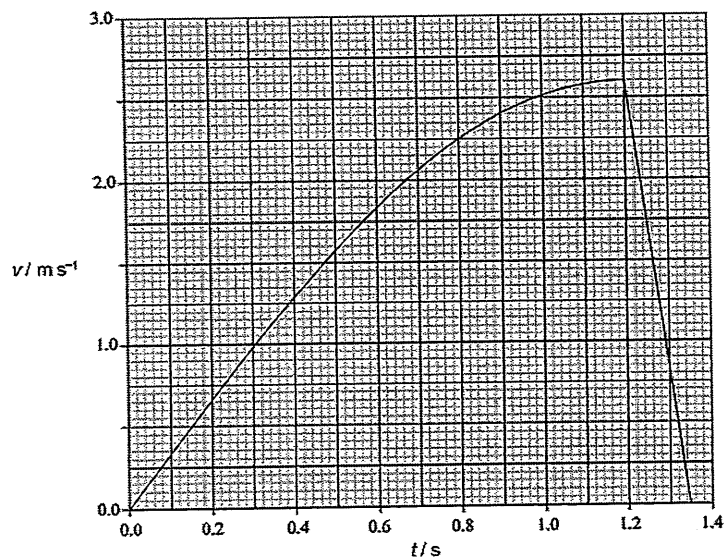
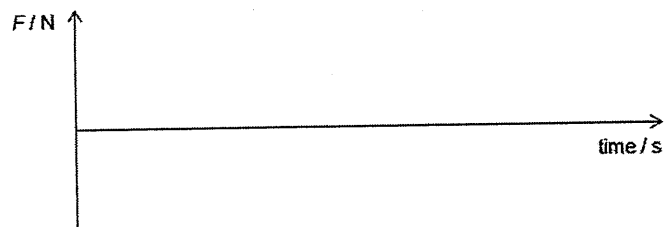


Fig. 2.2

- a. Just before the ball hits the wall,
 - i. explain why the tension is not equal to the weight of the ball. [2]
 - ii. with reference to Fig. 2.2, estimate the tension in the cable. [3]
- b. Taking rightwards as positive, when the ball collides with the wall,
 - i. determine the change in momentum of the ball. [3]
 - ii. On the axes below, sketch a clearly labelled graph to show the variation with time of the force F on the ball during the collision. [2]



- c. i. State the principle of conservation of momentum. [1]
- ii. The ball has lost momentum in its collision with the wall. Explain whether the principle of conservation of momentum is violated in this situation. [2]

[Total: 13]

3. [NJC/Prelims 2014/P2/3]

- a. State the Principle of Conservation of Momentum. [1]
- b. A billiard ball, A collides head-on with stationary billiard ball B of mass 150 g as in Fig. 3.1.



Fig. 3.1

The force F exerted on ball B by ball A varies with time t as shown in Fig. 3.2.

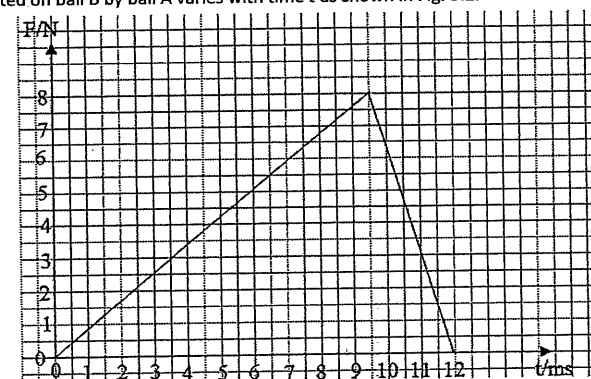


Fig. 3.2

- i. Calculate the speed of ball B after the collision. [2]
- ii. Explain, using Newton's Third law of Motion, the relationship between the impulse experienced by A and the impulse experienced by B during impact. [2]
- c. Ball B travelled towards the edge of the table and rolls off the edge with negligible speed. A net is placed 1.0 m below the table to catch the ball. Ball B lands on the net and is brought to a complete stop in a time of 0.50 s. Calculate the average force exerted on the net by ball B. [3]

[Total: 8]

4. [NYJC/Prelims 2014/P2/2]

- a. State the principle of conservation of momentum. [1]
- b. A 0.150 kg toy helicopter is moving at a constant altitude of 75.0 m with a speed of 1.50 m s^{-1} when a shooter fires vertically up and hits it as shown in Fig. 4.1. The mass of the bullet is 1.00 g and the initial speed of the bullet is 100 m s^{-1} .

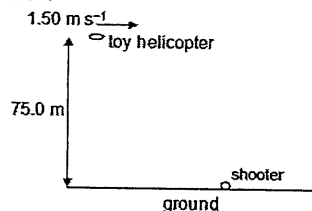


Fig. 4.1

Assume negligible air resistance and ignore height of shooter, determine

- the distance of the toy helicopter from the shooter when he shoots [3]
- the momentum of the toy helicopter and bullet immediately after the hit. Assume that the bullet is embedded in the toy helicopter after the hit. [2]

[Total: 6]

5. [PJC/Prelims 2014/P3/1]

Fig. 5.1 shows a box of mass 5.0 kg connected to a hanging mass of 2.0 kg by a light rope over a frictionless pulley. The box is then released from rest and the frictional force acting between the moving box and the slope is 4.0 N.

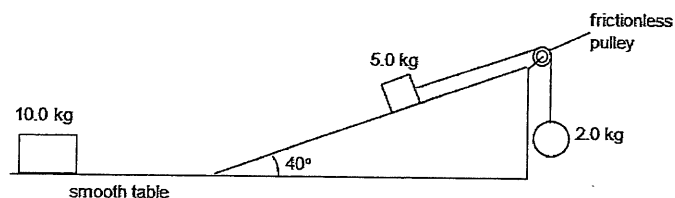


Fig. 5.1

- Calculate the magnitude of acceleration of the box. [3]
- The box took 3.0 s to reach the bottom of the slope. At that instance, the rope snapped and the box continues its motion along the smooth table without any changes to its speed. It then collides with a block of mass 10.0 kg, causing the 10.0 kg block to move to the left with a speed of 2.0 m s^{-1} .
 - Determine the speed of the 5.0 kg box immediately after collision. [2]
 - Discuss whether momentum is still conserved in the above collision if the table is rough instead. [2]

[Total: 7]

6. [RVHS/Prelims 2014/P3/6]

A toy manufacturer is testing out a newly designed toy gun which will be launched before the end-year festive season.

In order to simulate a user holding the gun, the gun is supported by strings at three points as shown in Fig. 6.1. An upward force of 2.5 N acts on point A, a force of 3.1 N acts on point B at an angle 10° to the vertical, and a force of 1.5 N acts on point C at an angle 21° to the vertical. The other ends of the strings are attached to a rigid horizontal structure.

Point A, C and the centre of gravity of the gun are assumed to lie on the same horizontal plane as the barrel. Point B is 5.0 cm below this horizontal plane.

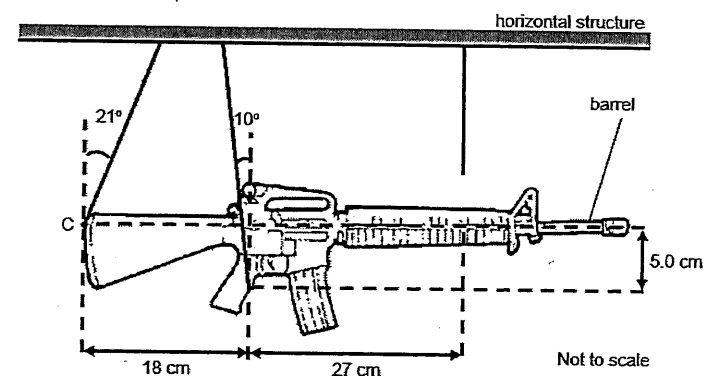


Fig. 6.1

- Show that the weight of the gun is 7.0 N. [1]
 - Calculate the distance of the centre of gravity of the gun from point C. [2]
 - State and explain how the gun will rotate when the contact forces at A and C are removed at the same time. [2]
- b. The user now loads and fires a rubber bullet horizontally to experience the recoil of the gun. The gun of effective barrel length of 40 cm fires a rubber bullet of mass 40 g with an exit velocity of 12 m s^{-1} . Assume the acceleration of the bullet as it travels down the barrel is constant. Calculate
- the momentum of the bullet as it leaves the barrel [2]
 - the time it takes to accelerate along the barrel [2]
 - the average force causing the acceleration [2]
 - the recoiling force of the gun on a user. [1]

- c. The recoiling force of the gun is found to be too large and causes discomfort to the user. Hence, a rubber pad is fitted to the butt of the gun as shown in Fig. 6.2.

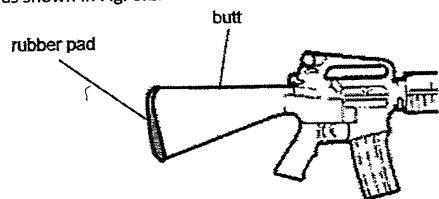


Fig. 6.2

Describe and explain how this pad will affect the recoil of the gun, as experienced by the user. [3]

- d. The rubber bullet has a head-on collision with a block of mass 85 g which is resting on a smooth platform as shown in Fig. 6.3. The bullet rebounds at a speed of 1.0 m s^{-1} after its impact with the block. The block slides towards the wall and causes a maximum compression of 15 cm on a spring that is attached to the wall.

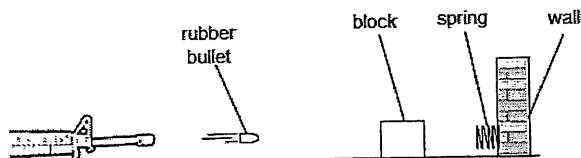


Fig. 6.3

- If the velocity of the bullet just before it hits the block is the same as its exit velocity from the barrel, determine the velocity of the block after its impact with the bullet. [2]
- Calculate the force constant of the spring. [3]

[Total: 20]

7. [SRJC/Prelims 2014/P2/2]

- a. Block A of mass 2.5 kg is resting on a rough table as shown in Fig. 7.1. The static frictional force acting on Block A by the table is 5.0 N. Block B of mass 4.0 kg is attached to block A via an inelastic string which passes over a smooth pulley. The other end of block A is held by an inelastic string attached to a wall while the other end of block B is attached to a spring that is being compressed by 2.0 cm. The spring constant is 1000 N m^{-1} .

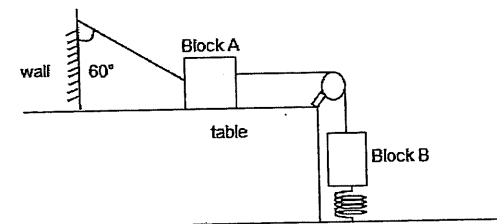


Fig. 7.1

- Show that the tension in the string between block A and block B is 19.2 N. [1]
 - Hence or otherwise, calculate the normal contact force exerted by the table on block A. [3]
- b. Object X of mass 8.0 kg is moving towards the right with a speed of 1.5 m s^{-1} while object Y of mass 10.0 kg is moving behind it in the same direction. The relative speed of approach between the objects is 1.0 m s^{-1} . At 1.5 s, object Y caught up with object X and both objects collide.

Fig. 7.2 shows a graph of momentum against time for object X.

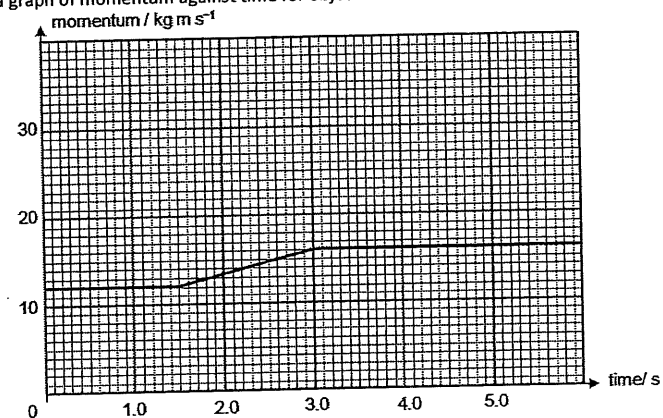


Fig. 7.2

- Calculate the final momentum of object Y. [2]
- Draw the momentum against time graph for colliding object Y in Fig. 7.2. [1]

[Total: 7]

8. [TJC/Prelims 2014/P3/1]*

A spring is placed on a flat surface and different weights are placed on it, as shown in Fig. 8.1.

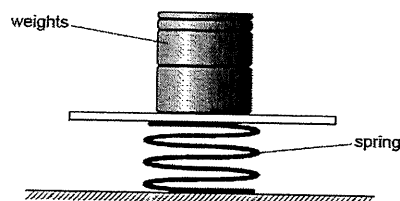


Fig. 8.1

The variation with weight of the compression of the spring is shown in Fig. 8.2.

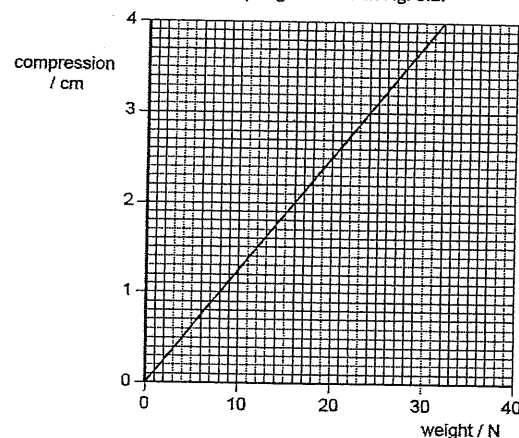


Fig. 8.2

The elastic limit of the spring has not been exceeded.

- a. i. Determine the spring constant k of the spring. [2]
- ii. Calculate the amount of elastic potential energy that is stored in the spring for a compression of 3.5 cm. [2]

- b. Two trolleys of masses 800 g and 2400 g are free to move on a horizontal table. The spring in (a) is placed between the trolleys and the trolleys are tied together using thread so that the compression of the spring is 3.5 cm, as shown in Fig. 8.3.

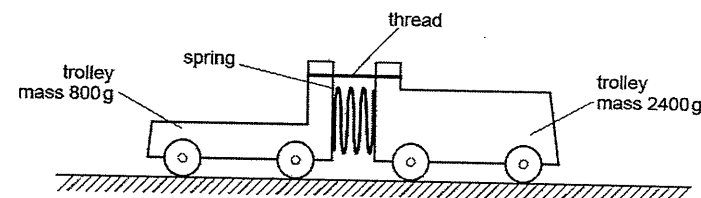


Fig. 8.3

Initially, the trolleys are not moving.

The thread is then cut and the trolleys move apart.

- i. Determine the ratio of $\frac{\text{speed of trolley of mass 800 g}}{\text{speed of trolley of mass 2400 g}}$ [2]
- ii. Use the answers in a. ii. and b. i. to calculate the speed of the trolley of mass 800 g. [3]
- c. In a separate experiment, a trolley of mass 850 g is held between two fixed points by means of two identical springs in a. as shown in Fig. 8.4.

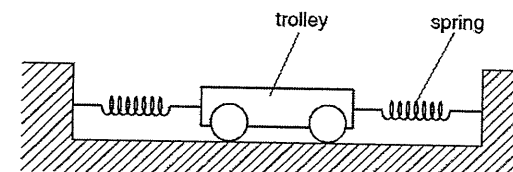


Fig. 8.4

When the trolley is in equilibrium, the springs are each extended by 4.5 cm.

The trolley is moved a distance of 1.5 cm along the direction of the springs. This causes the extension of one spring to be increased and the extension of the other spring to be decreased. The trolley is then released. The trolley accelerates and reaches its maximum speed at the equilibrium position.

Determine the maximum speed of the trolley.

[3]

[Total: 12]

9. [TPJC/Prelims 2014/P3/1]

- a. Three blocks of masses 0.40 kg, 1.00 kg and 0.30 kg move on a frictionless horizontal track with speeds of 5.0 m s^{-1} , 3.0 m s^{-1} and 4.0 m s^{-1} respectively as shown in Fig. 9.1. Velcro couplers make the blocks stick together after colliding, moving off as a train.

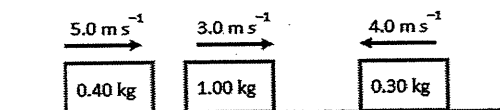


Fig. 9.1

Show that the final speed of the train of three blocks immediately after colliding is 2.24 m s^{-1} . [2]

- b. The train of three blocks from part a., then makes a head-on elastic collision with another block B of mass 5.0 kg, which has a spring attached and is suspended by a stiff light rod pivoted at a point C, as shown in Fig. 9.2. The spring on block B will be compressed during the collision, then extended, before the train and block B separate.

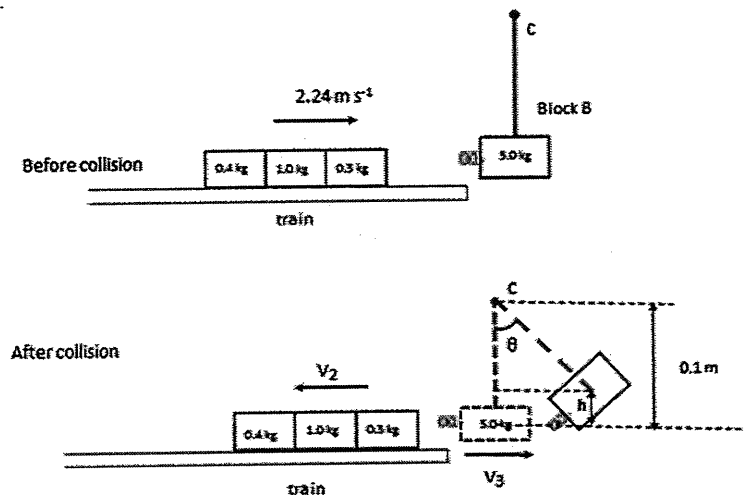


Fig. 9.2

- Calculate v_3 , the speed of block B after the collision. [3]
- The distance of block B's centre to pivot point C is 0.10 m (as shown in Fig. 9.2). Calculate the angle θ subtended by the rod when block B reaches its maximum height. [2]

[Total: 7]

10. [VIC/Prelims 2014/P3/1]

Fig. 10.1 shows the momentum against time graphs for two colliding lorries A and B.

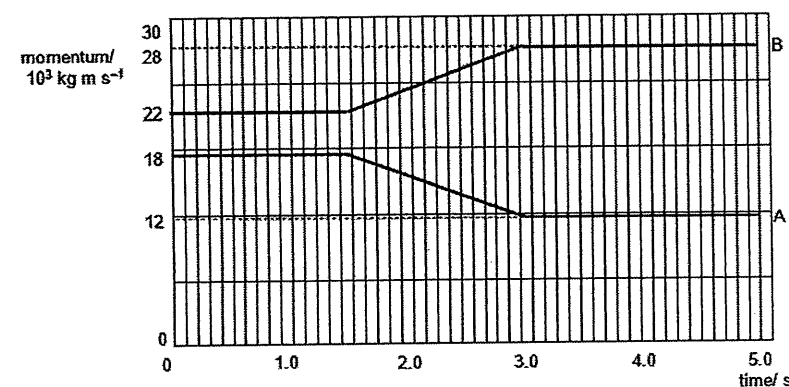


Fig. 10.1

The masses of lorries A and B are $2.0 \times 10^3 \text{ kg}$ and $4.0 \times 10^3 \text{ kg}$ respectively.

- State what the gradients of the graphs during the collision represent. [1]
- Explain why the gradients of the graphs during the collision have opposite signs. [1]
- Calculate the force acting on lorry B during the collision. Hence, or otherwise, calculate the force acting on lorry A during the collision. [2]
- Use Newton's laws of motion to explain whether momentum is conserved in this collision. [3]
- Calculate the change in the kinetic energy of the system and hence deduce whether the collision is elastic or inelastic. [3]

[Total: 10]

11. [YJC/Prelims 2014/P2/1]

Trolley A of mass 6.0 kg travelling at a speed of 5.0 m s^{-1} collides head-on with trolley B of mass 10 kg travelling at a speed of 3.0 m s^{-1} as shown in Fig. 11.1. Trolley B then travels backwards after the collision with a speed of 2.0 m s^{-1} .

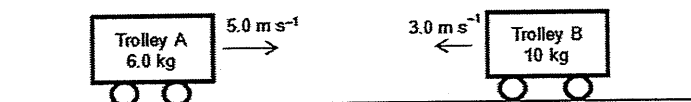


Fig. 11.1

- Determine the speed of trolley A after the collision. [2]

- b. Fig. 11.2 shows how the velocity of trolley B varies with time.
Sketch the corresponding velocity-time graph for trolley A.

[2]

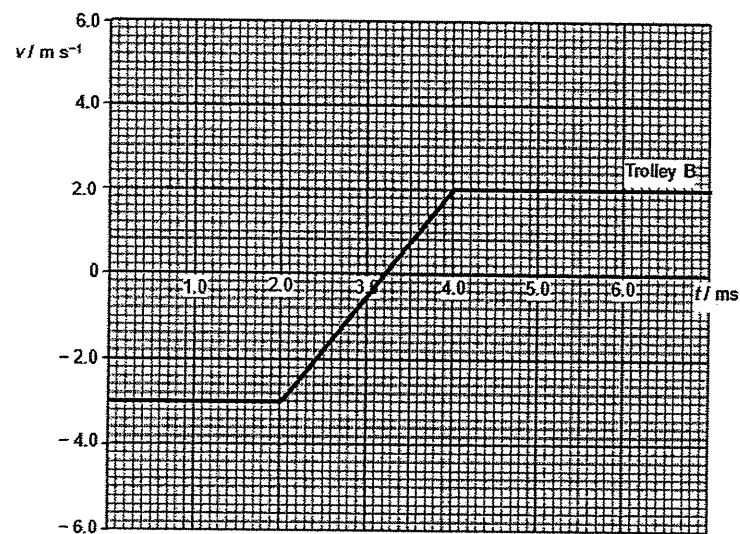


Fig. 11.2

- c. Explain whether the collision is an elastic or inelastic collision. Substantiate your conclusion with workings. [2]
- d. After trolley B collides with trolley A, it rolls down a rough slope as shown in Fig. 11.3. A spring is placed at the bottom of the slope and is attached to a wall. The spring is compressed by 5.0 cm before trolley B comes to complete rest momentarily. The spring has spring constant of 2000 N m^{-1} .

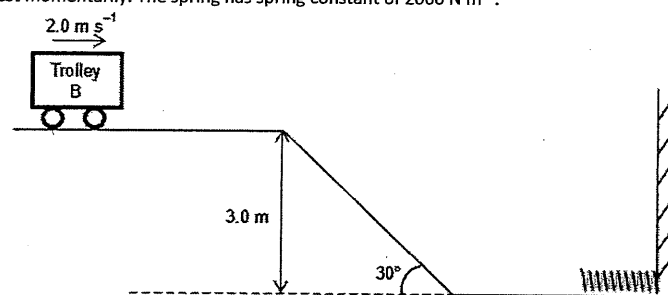


Fig. 11.3

- i. Calculate the average frictional force along the slope. [2]
- ii. State and explain the change to the compression of the spring if trolley B has a smaller mass. [2]

[Total: 10]

	Raffles Institution Physics Department Revision Package 2	Collisions	6
---	---	-------------------	----------

III. Solutions and Answers

The suggested solutions and answers below are only for your reference and have not been verified to be error-free. If you have any doubt, please discuss with your tutor to clarify.

MCQs

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
C	C	C	B	A	C	D	D	A	B	D	D	B	C	C

16	17
B	A

1 C $\Delta p = \text{area under } F - t \text{ graph} = \frac{1}{2}(10)(0.3) = 1.5 \text{ N s}$

2 C Mass of recoiling daughter nucleus $= 220u - 4u = 216u$

By the Conservation of momentum

$$0 = m_\alpha v_\alpha + m_D (-v_D)$$

$$\Rightarrow \frac{v_\alpha}{v_D} = \frac{m_D}{m_\alpha} = \frac{216}{4} = 54$$

$$\frac{E_\alpha}{E_D} = \frac{m_\alpha v_\alpha^2}{m_D v_D^2} = \frac{m_\alpha}{m_D} \times \left(\frac{m_D}{m_\alpha}\right)^2 = \frac{m_D}{m_\alpha}$$

$$\therefore E_D = E_\alpha \times \frac{m_\alpha}{m_D} = \frac{1}{54} E_\alpha$$

3 C Applying conservation of momentum to trolley.

$$m_T u_T = (m_T + m_S) V$$

$$2.0(0.500) = (2.0 + 0.500) V$$

$$V = 0.40 \text{ m s}^{-1}$$

$$\Delta p_T = m_T (v_T - u_T) = 2.0(0.40 - 0.50) = -0.20 \text{ N s}$$

Raffles Institution

Revision Package (H2 Physics)

4 B Let x be the recoiling nucleus.

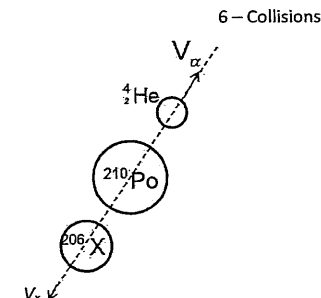
By conservation of momentum:

total initial momentum = total final momentum

$$0 = m_\alpha v_\alpha + m_x (-v_x) \Rightarrow \frac{v_\alpha}{v_x} = \frac{m_x}{m_\alpha}$$

$$\frac{E_x}{E_\alpha} = \frac{m_x v_x^2}{m_\alpha v_\alpha^2} = \frac{m_x}{m_\alpha} \times \left(\frac{m_\alpha}{m_x}\right)^2 = \frac{m_\alpha}{m_x}$$

$$\therefore E_x = E_\alpha \times \frac{m_\alpha}{m_x} = \frac{4}{210 - 4} E_\alpha = \frac{2}{103} E_\alpha$$



5 A By conservation of momentum: $m_A u_A + m_B (-u_B) = m_A v_A + m_B v_B$

Since it is an elastic collision, relative speed of approach will be equal to that of separation

$$u_A - (-u_B) = v_B - v_A$$

6 C By the Conservation of momentum, $0 = m_\alpha v_\alpha + m_D (-v_D) \Rightarrow \frac{v_\alpha}{v_D} = \frac{m_D}{m_\alpha}$

$$\frac{E_\alpha}{E_D} = \frac{m_\alpha v_\alpha^2}{m_D v_D^2} = \frac{m_\alpha}{m_D} \times \left(\frac{m_D}{m_\alpha}\right)^2 = \frac{m_D}{m_\alpha}$$

$$\therefore E_\alpha = E_D \times \frac{m_D}{m_\alpha} = 200 \times \frac{40}{4} = 2000 \text{ eV}$$

7 D By the Conservation of momentum

$$m_A u_A + m_B u_B = (m_A + m_B) V$$

$$(2.0)(3.0) + 0 = (2.0 + 1.0) V$$

$$V = 2.0 \text{ m s}^{-1}$$

$$\text{Loss in Kinetic energy} = \frac{1}{2} m_A u_A^2 - \frac{1}{2} (m_A + m_B) V^2 = \frac{1}{2} (2.0)(3.0)^2 - \frac{1}{2} (3.0)(2.0)^2 = 3.0 \text{ J}$$

8 D By the Conservation of momentum, taking towards the right as positive,

$$m_A u_A + m_B u_B = m_A v_A + m_B v_B$$

$$(4.0)(5.0) + (3.0)(2.0) = (4.0) v_A + (3.0) v_B \Rightarrow 4v_A + 3v_B = 26 \text{ --- [1]}$$

Since it was an elastic collision,

$$u_A - u_B = v_B - v_A$$

$$5.0 - 2.0 = v_B - v_A \Rightarrow v_B = 3 + v_A \text{ --- [2]}$$

$$\text{Solving for simultaneous equations: } v_A = 2.4 \text{ m s}^{-1}, v_B = 5.4 \text{ m s}^{-1}$$

- 9 A The additional momentum due to the sand in 1 sec = mv

$$\text{By Newton's 2nd Law, the force on the belt due to the sand} = \frac{mv}{1} = mv$$

In order to keep the belt at constant speed, an equal in magnitude and opposite force that must be exerted on the belt.

- 10 B At maximum height, the bomb is moves horizontally with velocity $100 \cos 40^\circ = 76.6 \text{ m s}^{-1}$
When the bomb explodes, momentum is conserved linearly. By the conservation of linear momentum, Horizontally:

$$m_i u_i = m_f v_f$$

$$m(76.6) = \left(\frac{1}{3}m\right)v_f$$

$$v_f = 230 \text{ m s}^{-1}$$

- 11 D By conservation of momentum, the two fragments must move in opposite directions such that the vector sum of their final momentum will cancel out.

$$Mv_1 = (3M)v_2 \Rightarrow \frac{v_1}{v_2} = 3$$

$$E = \frac{1}{2}Mv_1^2 + \frac{1}{2}(3M)v_2^2 = \frac{1}{2}Mv_1^2 + \frac{3}{2}M\left(\frac{v_1}{3}\right)^2 = \frac{1}{2}Mv_1^2 + \frac{1}{3}\left(\frac{1}{2}Mv_1^2\right) = \frac{4}{3}E_i$$

$$E_f = \frac{3}{4}E$$

- 12 D Options A and B cannot be true because momentum and energy are different quantities and cannot be converted into each other.

Option C is wrong because momentum of the system is conserved as there is no mention of external forces acting on the system.

- 13 B By the conservation of linear momentum, Horizontally, the momentum of the ball leaving the table has to be equal to the momentum of ball and the cart as it moves off together (inelastic collision).

$$m_i u_i = m_f v_f$$

$$0.300(10) = (0.300 + 1.2)v$$

$$3.0 = 1.5v$$

$$\therefore p_f = 3.0 \text{ kg m s}^{-1} \quad \& \quad v = 2.0 \text{ m s}^{-1}$$

- 14 C A is not possible as this means final momentum is mv to the right (ball must rebound along same path)
B is not possible as this means final momentum is zero.
C is possible as this means final momentum is mv to the left $\Rightarrow$ speed and hence KE has decreased.
D is not possible as this

- 15 C By the Conservation of momentum

$$m_A u_A + m_B u_B = (m_A + m_B)V$$

$$mu + 0 = (2m)V \Rightarrow V = 0.5u$$

$$\text{Loss in kinetic energy} = \frac{1}{2}mu^2 - \frac{1}{2}(2m)\left(\frac{1}{2}u\right)^2 = E - \frac{1}{2}E = \frac{1}{2}E$$

- 16 B Change in momentum, $\Delta p = \text{Area under } F-t \text{ graph} = \frac{1}{2}(2 \times 3) + (2 \times 3) = 9.0 \text{ kg m s}^{-1} = m\Delta v$

$$9.0 = (2.0)(v - 2.0)$$

$$v = 6.5 \text{ m s}^{-1}$$

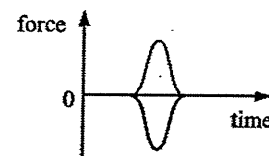
- 17 A By conservation of momentum, $0.040 \times 300 = (2.00 + 0.040)v \Rightarrow v = 5.882 \text{ m s}^{-1}$

$$\text{By conservation of energy, } \frac{1}{2}(2.00 + 0.040)(5.882)^2 = (2.00 + 0.040)gh \Rightarrow h = 1.76 \text{ m}$$

Structured Questions

1. [JJC/Prelims 2014/P2/2]

- a. i.



- ii. When a variable force acts on an object, the change in momentum of the object is given by the area under the force-time graph.

The answer to i. implies that the change in momentum of A and B are equal in magnitude and opposite in directions.

Consider A and B as one system, their changes in momentum cancel out vectorially thus the total momentum of the system remains unchanged.

- iii. By Conservation of Linear Momentum,

$$(2m + 3m)V_0 = (3m)\left(-\frac{1}{3}V_0\right) + 2mV_B \Rightarrow V_B = 3V_0$$

- iv. The internal explosion in the two-stage rocket gives additional forward force to component B, and hence its velocity increases.

Furthermore, with a decrease in mass after discarding component A which is used as a fuel module, a greater acceleration can be experienced with the same thrust.

- b. i. The object experiences three forces acting on it namely weight W , upthrust U and a viscous force R when it moves in the liquid which opposes its motion.

Net force on object,

$$W - U - R = ma$$

As the speed of object increases, viscous force increases,

acceleration of the object decreases until zero which implies that the object is moving at constant velocity.

Hence the speed increases to a constant terminal value.

- ii. since $v \propto a^2$

$$0.5 = k(2.0)^2 \text{ ---- (1)}$$

$$0.25 = k(a_2)^2 \text{ ---- (2)}$$

$$\frac{(1)}{(2)} \quad \frac{0.5}{0.25} = \frac{2.0^2}{a_2^2}$$

$$a_2 = 1.4 \text{ mm}$$

2. [MJC/Prelims 2014/P2/1]

- a. i. Since the ball moves along the arc of a circle, there will be a resultant force providing for the centripetal force. Hence the tension is larger than the weight.

- ii. From the graph, $v = 2.6 \text{ m s}^{-1}$ at 1.2 s

Using Newton's Second Law, just before the ball hits the wall, it is almost vertical

$$T - mg = mv^2/r$$

$$T - (380 \times 9.81) = 380 \times 2.6^2 / 5.8$$

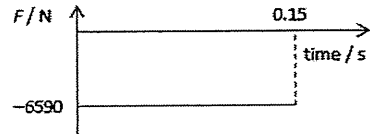
$$T = 4170 \text{ N}$$

- b. i. Change in momentum $= p_f - p_i = m(v_f - v_i) = 380(0 - 2.6) = -988 \text{ N s}$

The change in momentum of the ball is directed to the left, hence must have negative sign.

- ii. By Newton's second law, $F = \frac{dp}{dt}$

$$F = 988 / 0.15 \text{ (time interval read off from graph)} = 6590 \text{ N}$$



- c. i. See definitions in lecture notes.

- ii. Momentum is conserved:

The system is the ball, the wall and the Earth. The loss in momentum of the ball is equal to the gain in momentum of the wall and Earth.

OR

Momentum is not conserved:

There is an external force acting on the ball (system), so the ball's momentum is not conserved.

3. [NJC/Prelims 2014/P2/3]

- a. See definitions in lecture notes.

- b. i. Direction towards the right is positive since the force exerted by A on B is towards the right.

$$\text{Impulse} = \text{area under F-t graph} = 0.5 (12 \times 10^{-3}) \times 8 = 0.048 \text{ N s}$$

$$\text{change in momentum of ball B} = 0.1(v_f - 0)$$

$$0.048 = 0.15 v_f$$

$$v_f = 0.32 \text{ m s}^{-1}$$

- ii. Newton's third law states that if a body A exerts a force on body B, then body B exerts an equal and opposite force on body A.

Force on B (due to A) is equal in magnitude and opposite in direction to force (due to B)

time (of contact) (t) is same for both AND Impulse $= Ft$

impulse on A is equal in magnitude and opposite in direction to impulse on B

- c. Impulse on net by ball $= F_{\text{by ball}}(t)$

By Newton's third law, force on net by ball $= -$ Force on ball by net

During impact,

Force on ball by net $=$ contact force on ball by net $=$ weight of ball $+$ force to stop motion of ball, f_{stop}

Taking downwards as positive, where v_i is the speed of ball before it hits the net

Using $v^2 = u^2 + 2as$ (taking downwards as positive)

$$v_f^2 = 2(9.81)(1)$$

$$v_f = 4.43 \text{ m s}^{-1}$$

$$f_{\text{stop}} = \frac{m}{t}(v_f - v_i) = \frac{0.15}{0.5}(0 - v_i) = \frac{0.15}{0.5}(0 - 4.43) = -1.329 \text{ N, downwards} = 1.329 \text{ N, upwards}$$

$$\text{Force on net by ball} = 0.15(9.81) + 1.329 = 2.80 \text{ N, downwards}$$



4. [NYJC/Prelims 2014/P2/2]

- a. See definitions in lecture notes.

- b. i. Considering bullet, taking up as positive,

$$s_y = u_y t + \frac{1}{2} a_y t^2$$

$$75.0 = 100t + \frac{1}{2}(-9.81)t^2$$

$$t = 0.780 \text{ s or } t = 19.6 \text{ s}$$

Considering toy, taking right as positive,

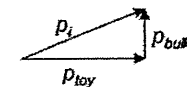
$$s_x = u_x t = 1.50(0.780) = 1.17 \text{ m}$$

$$\text{Distance} = \sqrt{s_x^2 + s_y^2} = \sqrt{(1.17)^2 + (75.0)^2} = 75.0 \text{ m}$$

- ii. Considering bullet, taking up as positive, $v_y = u_y + a_y t = 100 + (-9.81)(0.780) = 92.3 \text{ m s}^{-1}$

By Conservation of Momentum,

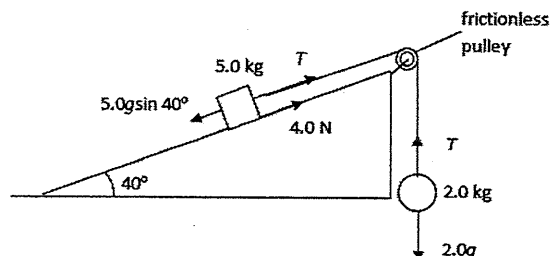
$$p_i = p_i = \overrightarrow{p_{\text{toy}}} + \overrightarrow{p_{\text{bullet}}}$$



$$\begin{aligned} p_i &= \sqrt{p_{\text{toy}}^2 + p_{\text{bullet}}^2} \\ &= \sqrt{(m_{\text{toy}} u_{\text{toy}})^2 + (m_{\text{bullet}} u_{\text{bullet}})^2} \\ &= \sqrt{[(0.150)(1.50)]^2 + [(1.0 \times 10^{-3})(92.3)]^2} \\ &= 0.243 \text{ kg m s}^{-1} \end{aligned}$$

5. [PJC/Prelims 2014/P3/1]

a.



By applying Newton's second law on both the masses and with acceleration down the slope, for the 2.0 kg mass:

$$T - 2.0g = 2.0a \quad (1)$$

for the 5.0 kg mass

$$5.0g \sin 40^\circ - 4.0 - T = 5.0a \quad (2)$$

Solving (1)+(2),

$$5.0g \sin 40^\circ - 4.0 - 2g = 7.0a$$

$$a = 1.13 \text{ m s}^{-2}$$

- b. i. Velocity of the ball just before collision, $u = at = (1.13)(3.0) = 3.39 \text{ m s}^{-1}$

Applying conservation of momentum and taking momentum to the left as positive,

$$(5.0)(3.39) = (10.0)(2.0) + (5.0)(v)$$

$$v = -0.61 \text{ m s}^{-1}$$

Therefore, the speed is 0.61 m s^{-1} .

- ii. Momentum is not conserved because with friction present, there will be a net external force.

6. [RVHS/Prelims 2014/P3/6]

- a. i. Weight of gun = $2.5 + 3.1 \cos 10 + 1.5 \cos 21 = 6.95 \text{ N} = 7.0 \text{ N}$

- ii. Taking moments about point C and let x be distance between C.G. and point C,

$$(7.0)(x) + (3.1 \sin 10)(5) = (2.5)(45) + (3.1 \cos 10)(18)$$

$$x = 23.5 \text{ cm}$$

- iii. By taking moments about point B, the weight provides a clockwise moment, and no moment due to force at B. Hence, the gun will rotate in a clockwise direction if both the forces at A and C are removed.

- b. i. momentum = $mv = (0.040)(12) = 0.48 \text{ kg m s}^{-1}$

- ii. time = distance / average speed = $0.4 / (0.50)(12) = 0.067 \text{ s}$

- iii. force = $dp/dt = (0.48 - 0) / 0.067 = 7.2 \text{ N}$

- iv. Using Newton's third law, the force on bullet by gun = force on gun by bullet = 7.2 N

- c. With the rubber pad, the change of momentum of the gun that acts on the user is unchanged.

However, the impulse is extended over a longer period of time.

This reduces the maximum force (average force) that the user experiences, and could reduce the discomfort that the user may experience using the gun.

OR

The rubber pad may convert Kinetic Energy to Elastic Potential Energy and Thermal Energy and reduces the kinetic energy of the gun on the user.

- d. i. Using principle of conservation of linear momentum

$$(0.040)(12) + 0 = (0.040)(-1) + (0.085)v$$

$$v = 6.12 \text{ m s}^{-1}$$

- ii. Loss in Kinetic Energy of block = Gain in Elastic Potential Energy of spring

$$\frac{1}{2} (0.085)(6.12)^2 = \frac{1}{2} k(0.15)^2$$

$$k = 142 \text{ N m}^{-1}$$

7. [SRJC/Prelims 2014/P2/2]

- a. i. By considering the vertical forces acting on block B,

As object is in equilibrium

$$T + F_{SB} = m_B g \quad (F_{SB} \text{ is the force by spring on block B})$$

$$T = 4(9.81) - 1000(0.02) = 19.24 \text{ N}$$

- ii. By considering the horizontal forces acting on block A,

$$F \cos 30 + f = T$$

$$F = (19.24 - 5) / \cos 30 = 16.44 \text{ N}$$

By considering the vertical forces acting on block A,

$$F \sin 30 + N = m_A g$$

$$N = 2.5(9.81) - 16.44 \sin 30$$

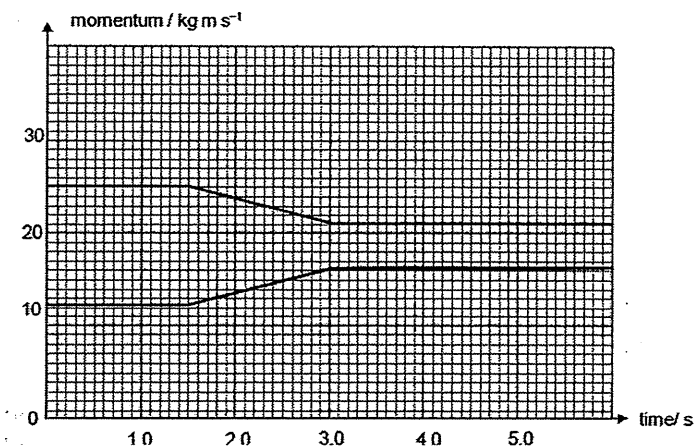
$$N = 16.3 \text{ N}$$

- b. i. Since relative speed of approach = $u_1 - u_2 = 1 \text{ m s}^{-1}$ and that the speed of object Y must be more than X, the initial speed of Y is 2.5 m s^{-1}

By conservation of momentum, $12 + (10)(2.5) = 16 + p_{fy}$

$$p_{fy} = 21 \text{ kg m s}^{-1}$$

- ii.



8. [TJC/Prelims 2014/P3/1]*

- a. i. k is the reciprocal of the gradient of the graph.
 $k = 32 / (4 \times 10^{-2}) = 800 \text{ N m}^{-1}$
- ii. EPE = area under the graph = $\frac{1}{2} \times 28 \times 3.5 \times 10^{-2} = 0.49 \text{ J}$
 OR
 $\text{EPE} = \frac{1}{2} kx^2 = \frac{1}{2} (800)(35 \times 10^{-2})^2 = 0.49 \text{ J}$
- b. i. Initial momentum = 0, hence both trolleys have equal and opposite momentum
 $mv = MV$
 $\frac{v}{V} = \frac{M}{m} = \frac{2400}{800} = 3.0$
- ii. EPE in spring = KE of trolleys
 $0.49 = \frac{1}{2} (800 \times 10^{-3})(v^2) + \frac{1}{2} (2400 \times 10^{-3})(\frac{1}{3} v^2)$
 $v = 0.96 \text{ m s}^{-1}$
- c. gain in energy of trolley = $\frac{1}{2} (800)(0.060^2 - 0.045^2) + \frac{1}{2} (800)(0.030^2 - 0.045^2) = 0.18 \text{ J}$
 $\frac{1}{2} (0.850)v^2 = 0.18$
 $v = 0.65 \text{ m s}^{-1}$

9. [TPJC/Prelims 2014/P3/1]

- a. By the conservation of momentum,
 $0.40(5.0) + 1.00(3.0) + 0.30(-4.0) = (0.40 + 1.00 + 0.30)v_1$
 $v_1 = 2.235 = 2.24 \text{ m s}^{-1}$
- b. i. relative speed of approach = relative speed of separation: $2.24 - 0 = v_3 - (-v_2)$ —(1)
 conservation of momentum: $1.7(2.24) = 1.7(-v_2) + 5.0(v_3)$ —(2)
 Solving, $v_3 = 1.14 \text{ m s}^{-1}$
- ii. conservation of energy: $5.0 \times g \times h = \frac{1}{2} (5.0)(1.136)^2$
 $h = 0.06577 \text{ m}$
 Looking at the right-angled triangle,
 $\cos \theta = (0.1 - 0.06577) / 0.1$
 $\theta = 70.0^\circ$

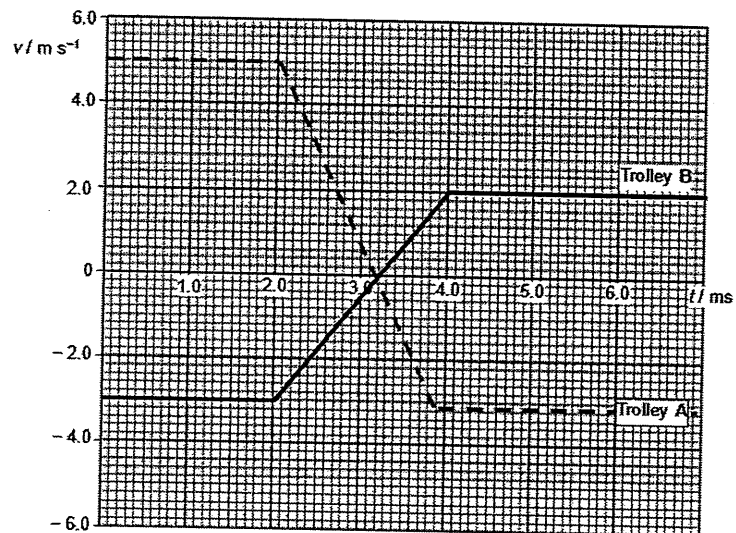
10. [VJC/Prelims 2014/P3/1]

- a. The gradient of the momentum-time graph represents the resultant force that acts on the object.
- b. According to Newton's third law, the resultant forces that acts on them are in opposite directions, therefore the gradients have opposite signs.
- c. $F_B = \text{gradient} = \frac{(28 - 22) \times 10^3}{1.5} = 4.0 \text{ kN}$
 By Newton's third law, $F_A = F_B = 4.0 \text{ kN}$
- d. According to Newton's second law, $F_A = \frac{dp}{dt} = \frac{m_A v_A - m_A u_A}{dt}$
 Similarly, $F_B = \frac{dp}{dt} = \frac{m_B v_B - m_B u_B}{dt}$
 According to Newton's third law, $F_A = -F_B$
 $\frac{m_A v_A - m_A u_A}{dt} = -\frac{(m_B v_B - m_B u_B)}{dt}$
 $m_A u_A + m_B u_B = m_A v_A + m_B v_B$
 Therefore, momentum is conserved since total initial momentum = total final momentum.
- e. Kinetic energy, $\text{KE} = \frac{1}{2} mv^2 = \frac{p^2}{2m}$
 Total initial KE, $= \frac{18000^2}{2(2000)} + \frac{22000^2}{2(4000)} = 1.42 \times 10^5 \text{ J}$
 Total final KE, $= \frac{12000^2}{2(2000)} + \frac{28000^2}{2(4000)} = 1.34 \times 10^5 \text{ J}$
 Change in KE, $\text{KE}_f - \text{KE}_i = (1.34 - 1.42) \times 10^5 = -8.0 \text{ kJ}$
 Therefore, collision is inelastic since KE is not conserved.

11. [YJC/Prelims 2014/P2/1]

- a. By conservation of momentum,
 $m_A u_A + m_B u_B = m_A v_A + m_B v_B$
 $6.0(5.0) + 10(-3.0) = 6.0(v_A) + 10(2.0)$
 $v_A = -3.3 \text{ m s}^{-1}$
 speed = 3.3 m s^{-1}

b.



c. KE before: $KE_i = \frac{1}{2}(6)(5)^2 + \frac{1}{2}(10)(3)^2 = 120 \text{ J}$

KE after: $KE_f = \frac{1}{2}(6)(3.3)^2 + \frac{1}{2}(10)(2)^2 = 53 \text{ J}$

Since there is loss in kinetic energy during the collision, the collision is inelastic.

d. i. $K.E_i + G.P.E_i + E.P.E_i = K.E_f + G.P.E_f + E.P.E_f + \text{work done against friction}$

$$\frac{1}{2}(10)(2)^2 + (10)(9.81)(3.0) + 0 = 0 + 0 + \frac{1}{2}(2000)(5 \times 10^{-2})^2 + (F_r)\left(\frac{3.0}{\sin 30^\circ}\right)$$

$$F_r = 52.0 \text{ N}$$

ii. The compression of the spring will be smaller.

Loss in the G.P.E and K.E is converted to gain in E.P.E. Since the mass is smaller, the loss in G.P.E and K.E will be smaller and the gain in E.P.E will be smaller